

EX NO: 5

IT22030 ETHICAL HACKING LABORATORY

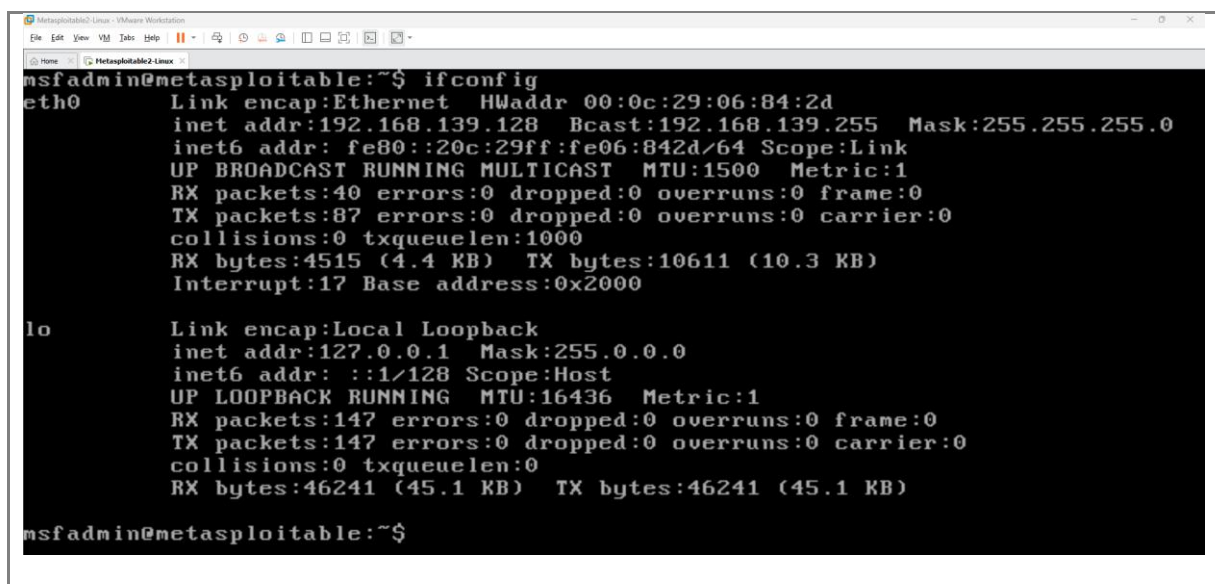
DATE:

Penetration Testing using Metasploit and Metasploitable

AIM: To perform a basic penetration test using the Metasploit Framework against the Metasploitable 2 vulnerable virtual machine, by scanning open ports with Nmap and exploiting the vsftpd 2.3.4 backdoor.

PROCEDURE:

1. Download Metasploitable 2 (a deliberately vulnerable Linux VM) from SourceForge and import it into VMware Workstation Player. Power on the VM and login with msfadmin / msfadmin. Note the IP address by running 'ifconfig'.



```
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 00:0c:29:06:84:2d
          inet addr:192.168.139.128  Bcast:192.168.139.255  Mask:255.255.255.0
          inet6 addr: fe80::20c:29ff:fe06:842d/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:40 errors:0 dropped:0 overruns:0 frame:0
          TX packets:87 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:4515 (4.4 KB)  TX bytes:10611 (10.3 KB)
          Interrupt:17 Base address:0x2000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:147 errors:0 dropped:0 overruns:0 frame:0
          TX packets:147 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:46241 (45.1 KB)  TX bytes:46241 (45.1 KB)

msfadmin@metasploitable:~$
```

2. On the Windows host, install Nmap from <https://nmap.org> and run 'nmap -sV <Metasploitable_IP>' to discover open ports and service versions.

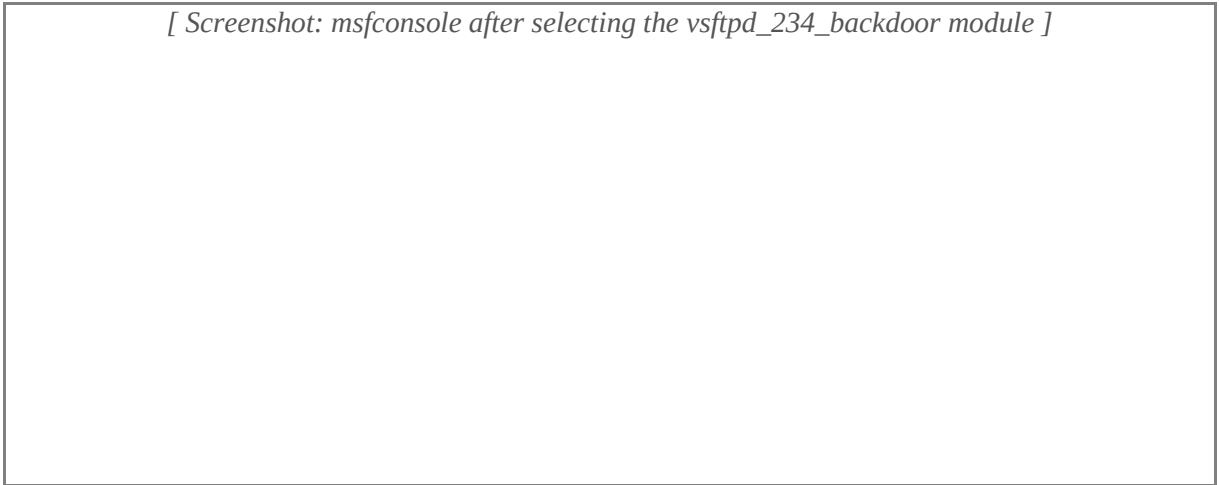
3. Install the Metasploit Framework on Windows from <https://www.metasploit.com/download>. Open msfconsole.

[Screenshot: msfconsole banner after launching Metasploit]



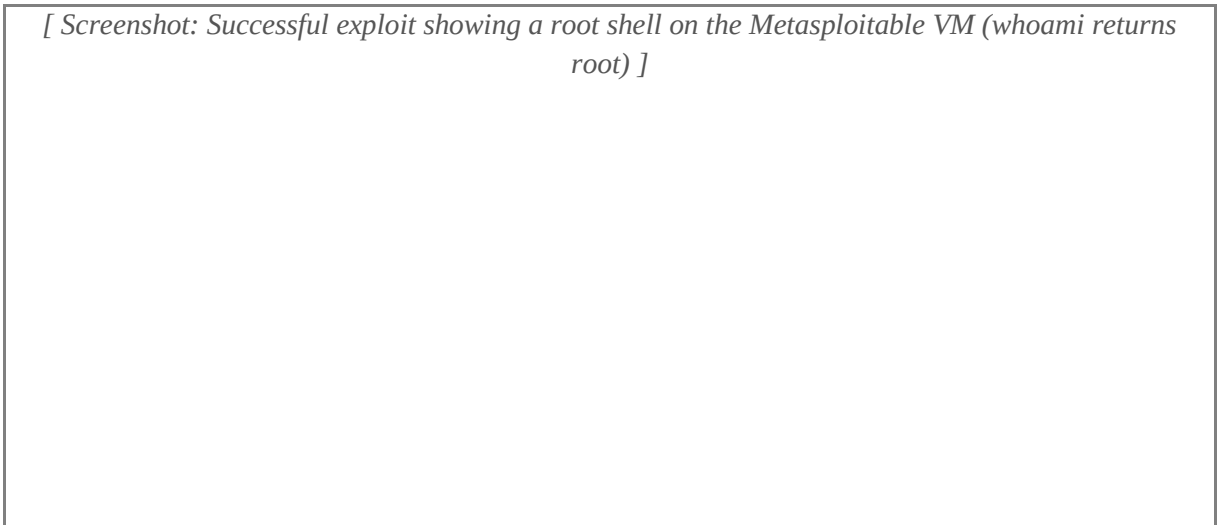
4. In msfconsole, type 'use exploit/unix/ftp/vsftpd_234_backdoor' to select the vsftpd backdoor exploit module.

[Screenshot: msfconsole after selecting the vsftpd_234_backdoor module]



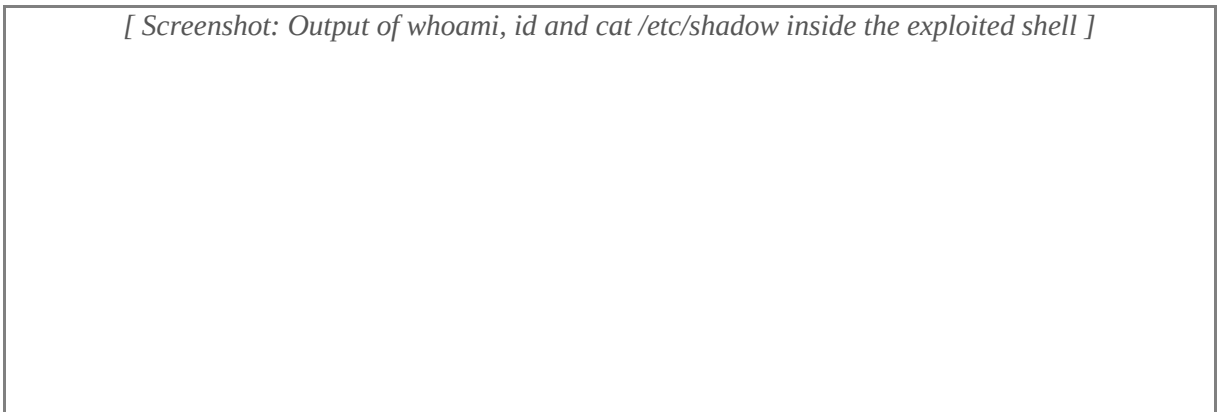
5. Set the target with 'set RHOSTS <Metasploitable_IP>' and run the exploit using 'exploit'. A root shell is returned by the vulnerable FTP service.

[Screenshot: Successful exploit showing a root shell on the Metasploitable VM (whoami returns root)]



6. Inside the obtained shell, run commands such as 'whoami', 'id' and 'cat /etc/shadow' to confirm full root access, then exit with 'exit'.

[Screenshot: Output of whoami, id and cat /etc/shadow inside the exploited shell]





RESULT: Thus, a basic penetration test was performed successfully using the Metasploit Framework against the Metasploitable 2 VM, and root access was obtained through the vsftpd 2.3.4 backdoor exploit.